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Substrate anchoring device

Abstract

A fastener for attaching together layers of a substrate. The fastener includes a slide body, a bolt, a conical spring, and a toggle having a pivot coupled with one end of the slide body. The conical spring is disposed over a shaft of the bolt and the shaft is engaged within the slide body. The slide body and toggle are inserted into an aperture through the layers of substrate in a first configuration until the conical spring contacts a front face of the layers of substrate. Compression of the conical spring against the front face advances the toggle a clearance length out of the aperture on a back face of the layers of a substrate such that the toggle can switch to a second configuration that prevents retraction through the aperture. Extension of the conical spring automatically engages the toggle with the back face of the substrate to prevent rotation of the slide body within the aperture.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

(1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference herein in its entirety and made a part of the present disclosure.

BACKGROUND

Field

(2) The disclosure generally relates to devices for attaching a plurality of substrates and more particularly to devices for attaching a plurality of substrates where only one side of a first substrate is accessible.

Related Art

(3) The disclosed embodiments were conceived in relation to the construction arts and in particular in relation to the addition of wooden decks to houses. Decks are extremely popular on new homes and are becoming extremely popular as add-ons to older homes. The National Association of Home Builders estimates that wood decks are included in nearly one-third of new homes, and are the most popular do-it-yourself construction project. A vast number of decks built in the last 30 years are in need of re-building, and many of these need to be redesigned with improved technology to make them safer. Deck collapses are not uncommon, and most deck collapses can be attributed to failure of the anchoring system anchoring the deck to the associated structure.

(4) A common method for attaching a deck to an associated structure is nailing a deck beam to the rim joists of the structure. This method, however, is prone to failure over time. While nails generally have sufficient tensile strength to resist the vertical shearing force created by the weight of the deck and its contents, nails do not provide great strength laterally and are prone to pullout. Wood screws and lag screws can be used in place of nails to provide additional lateral strength, however screws are also prone to failure, particularly when there is localized rot or other decay proximate to the screw, or aging of the substrate causes enlargement of the hole around the screw.

(5) A superior method for attaching a deck to an associated structure utilizes a bolt and nut assembly. Generally this method involves placing a bolt fitted with a washer into a hole which has been pre-drilled through both the deck beam and the rim-joist. The bolt is then fitted with a second washer and nut on the back side of the rim joist. The nut and second washer provide substantial resistance against pull-out forces even when there is localized rot common to older decks. Nevertheless, despite the superiority of the bolt and nut assembly, lack of access to the backside of the rim joist in many applications makes the use of the bolt and nut assembly impractical or even impossible.

(6) Many attempts have been made to provide a device for joining two or more substrates which device provides high pull-out resistance and which can be deployed when the substrates can be accessed only from one side. For example, in U.S. Pat. No. 3,017,800, Cohen teaches an anchor bolt with an expanding sleeve particularly suited for joining two asbestos panels. The anchor bolt taught by Cohen however, requires a bolt-sleeve made from a compressible, flexible material which

makes it ill-suited for heavy-duty applications such as joining a deck to a home.

(7) In U.S. Pat. No. 8,439,613, Prentice teaches a fastener apparatus that features projecting locking members that extend and retract when a central drive shaft is rotated. To use this fastener, however, an internal shaft must be rotated to extend locking members before the fastener apparatus may be tightened.

(8) In U.S. Pat. No. 3,241,420, Passer La Roy teaches a hollow wall fastening receptacle which can be used in combination with a screw to secure an object to a wall. This fastening receptacle requires that a hole be drilled through the wall for deployment of the device before the object can be attached to the fastening receptacle via a screw.

(9) Thus, there is a need for a durable device for easily attaching a plurality of substrates together when only one side of the substrates is accessible. It is therefore, an object of one or more of the embodiments disclosed herein to provide a fastening device for joining substrates which may be deployed when only one side of the substrates is accessible.

(10) A further object of one or more of the embodiments disclosed herein is to provide a fastening device which can be secured using a single tool.

(11) A further object of one or more of the embodiments disclosed herein is to provide a fastening device which can be deployed through a substrate without pre-drilling a hole through the substrates.

SUMMARY

(12) It is an object of one or more of the embodiments disclosed herein to provide a fastening device for joining a plurality of substrates which provides for high pull-out resistance, and simplicity of use. In one configuration, an internally threaded shoulder bolt is used in combination with one or more (e.g., a plurality of) deployable stop bars. The deployable stop bars are hingedly attached to a slide collar which is slidably mounted on a slide bolt. The slide bolt is threaded to work in conjunction with the internal threads of the shoulder bolt. In some configurations, a spring is provided to bias the stop bars to positions which are substantially perpendicular to the slide bolt. When the fastening device is in its undeployed state, a retaining ring is mounted on the slide collar to hold the stop bars in positions where each stop bar is substantially parallel to the slide bolt.

(13) To secure the fastening device, the fastening device is driven through the substrates to be joined. The fastening device may be driven through pre-drilled holes in the substrates. In one embodiment the fastening device has a rigid and pointed tip and may be pushed or hammered through the substrates. In another embodiment the fastening device has a drill tip and may be driven through the substrates by rotation of the fastening device. As the fastening device is driven through the substrates, the retaining ring is pressed against the first substrate and is urged off of the slide collar, allowing the spring to bias the stop bars towards a position perpendicular to the slide bolt. When the slide collar and stop bars emerge on the opposite side of the substrates, the stop bars are deployed. Once the stop bars are deployed, the shoulder bolt is rotated causing the slide bolt to move inwardly into the hollow sleeve of the shoulder bolt and compressing the substrates between the stop bars and a flange on the shoulder bolt. The shoulder bolt is rotated until a desired torque is reached.

(14) In one embodiment, a fastener for attaching together layers of a substrate, includes a slide body configured to be received within an aperture through the substrate. The slide body has a first end and a second end. The first end has a tapped opening. A bolt engages with the slide body. The bolt has a head and a threaded shaft. The threaded shaft is engageable within the tapped opening of the slide body. A toggle prevents removal of the fastener from the aperture. The toggle has a pivot coupled with the second end of the slide body. The toggle has a first configuration in which the toggle can advance with within the aperture and a second configuration which prevents the fastener from being retracted from the aperture. A conical spring has a narrow end and a wide end. The conical spring is disposed over the threaded shaft of the bolt with the narrow end of the conical spring adjacent to the head of the bolt and the wide end of the conical spring adjacent to the first

end of the slide body. The slide body is inserted within the aperture from a front face of the substrate with the toggle in the first configuration and advanced within the aperture until the wide end of the conical spring engages with the front face of the substrate. Compression of the conical spring further advances the slide body within the aperture and advances the toggle a clearance length out of the aperture on a back face of the substrate such that the toggle can switch to the second configuration. Extension of the conical spring retracts the slide body within the aperture and engages the toggle with the back face of the substrate to prevent retraction of the slide body through the aperture. The bolt is tightened to compress the conical spring and to engage with the front face through the conical spring.

(15) In another embodiment, the slide body is cylindrical and has an outer diameter between 0.125" and 1.0" and the toggle in the first configuration has an outer diameter that is smaller than or equal to a diameter of the aperture of the substrate and a diameter that is larger than the diameter of the aperture in the second configuration.

(16) In another embodiment, the toggle has a single stop bar, the single stop bar pivotably engaged with the second end of the slide body by a pivot.

(17) In another embodiment, the toggle is eccentrically mounted with respect to the pivot such that the force of gravity can rotate the single stop bar about the pivot into the second configuration.

(18) In another embodiment, the toggle has a pair of stop bars that rotate about a pivot and a torsional spring biases the toggle into the second configuration.

(19) In another embodiment, the conical spring is compressible into a flattened state.

(20) In another embodiment, the fastener includes a washer. The washer is disposed between the wide end of the conical spring and the slide body. The washer provides a bearing surface for the conical spring to act against the front face of the substrate.

(21) In another embodiment, a compressive length of the conical spring is the same as or greater than the clearance length of the toggle from the aperture.

(22) In another embodiment, a length of the fastener from the first end of the slide body to the pivot of the toggle is equal to the sum of a width of each of the layers of the substrate, plus the clearance length of the toggle, minus a fraction of the width of one of the plurality of substrates, the fraction between selected from between $\frac{1}{8}$ and $\frac{7}{8}$.

(23) One embodiment of a fastener for attaching together layers of a substrate through an aperture includes a slide body having a first end and a second end, the first end having a tapped opening. A bolt has a head and a threaded shaft. A toggle has a pivot coupled with the second end of the slide body; the toggle has a first configuration in which the toggle has a first diameter and a second configuration in which the toggle has a second diameter. A conical spring is disposed on the shaft between the slide body and the head. The toggle in the first configuration and the slide body are inserted within the aperture from a front face of the substrate and compression of the conical spring against the front face advances the toggle a clearance length out of the aperture on a back face of the substrate such that the toggle can switch to the second configuration. Extension of the conical spring automatically engages the toggle with the back face of the substrate and the bolt is tightened to compress the conical spring between the head and the front face.

(24) In another embodiment, the toggle has a single stop bar, the single stop bar pivotably engaged with the second end of the slide body by a pivot.

(25) In another embodiment, the toggle is eccentrically mounted with respect to the pivot such that the force of gravity can rotate the single stop bar about the pivot into the second configuration.

(26) In another embodiment, the toggle comprises a pair of stop bars that rotate about a pivot and a torsional spring biases the toggle into the second configuration.

(27) In another embodiment, the conical spring is compressible into a flattened state.

(28) In another embodiment, the fastener includes a washer disposed between a wide end of the conical spring and the slide body. The washer provides a bearing surface for the conical spring to act against the front face of the substrate.

(29) In another embodiment, a compressive length of the conical spring is the same as or greater than the clearance length of the toggle from the aperture.

(30) Embodiments of the disclosure possess many other advantages, and has other objects which may be made more clearly apparent from a consideration of several embodiments of the invention. For this purpose, there are shown a few forms in the drawings accompanying and forming part of the present specification, and which drawings, unless as otherwise indicated, are true scale. These forms will now be described in detail, illustrating the general principles of the embodiments; but it is to be understood that this detailed description is not to be taken in a limiting sense, since the scope of the invention is best defined by the appended claims.

Description

BRIEF DESCRIPTION

(1) In the accompanying drawings in which is shown several of the various possible embodiments of the subject anchoring device.

(2) FIG. 1 is a perspective of the fastening device in its insertion position and fastening three substrates.

(3) FIG. 2 is a perspective of the fastening device in its undeployed state.

(4) FIG. 3 is an exploded view of the fastening device.

(5) FIG. 4 is a cross-section of the fastening device fastening three substrates.

(6) FIG. 5 is a cross section of the anchor portion of the fastening device engaged in the shoulder bolt portion.

(7) FIG. 6 shows another embodiment of a fastener.

(8) FIG. 7 shows the fastener of FIG. 6 within an aperture of a substrate.

(9) FIG. 8 shows the fastener of FIG. 6 advanced within the aperture.

(10) FIG. 9 shows a toggle of the fastener of FIG. 6 engaged with a back face of the substrate.

(11) FIG. 10 shows the fastener of FIG. 6 in a fully tightened configuration.

(12) FIG. 11 shows a sectional view of the fastener of FIG. 6 in the fully tightened configuration.

(13) FIG. 12 shows yet another embodiment of a fastener.

DETAILED DESCRIPTION

(14) Referring initially to FIGS. 2 and 3, a fastening device in accordance with the present invention is shown. The fastening device includes generally a shoulder bolt 10 and an anchor 20. The shoulder bolt has an internally threaded shaft 11 and a head 12, the diameter of the head being larger than the diameter of the shaft.

(15) The anchor 20 comprises a slide bolt 21 which is threaded on a first end to work in conjunction with the internally threaded shaft 11. A second end of the slide bolt defines a plurality of slide bolt notches 26. A slide collar 22 is slidably disposed on a second end of the slide bolt 21 and a cap 23 is fixedly attached to the second end of the slide bolt 21. A biasing element, such as a spring 24, is disposed on the second end of the slide bolt between the cap 23 and the slide collar 22.

(16) The anchor 20 further comprises a plurality of stop bars 25, hingedly attached to the slide collar 22. In a preferred embodiment, hinge pins 30 are utilized to hingedly attach the stop bars to the slide collar. The stop bars 25 are shaped to define substantially V-shaped projections 27 which are sized to fit within the slide bolt notches 26 defined in the second end of the slide bolt 21. When the stop bars 25 are disposed so that the projections 27 are within the slide bolt notches 26, the slide collar and the stop bars define a cylinder with an outer diameter substantially the same as the outer diameter of the internally threaded shaft 11.

(17) The slide collar 22 defines a retaining ring groove 29 and each of the stop bars also define stop bar grooves 30 such that when the stop bars 25 are disposed so that the projections 27 are within the slide bolt notches 26 the grooves defined by the stop bars are aligned with the retaining ring

groove **29** defined by slide collar **22**. When the fastening device is in its undeployed state, a retaining ring **28** is fitted within the retaining ring groove **29** and the stop bar grooves **30** to retain the projections **27** within the slide bolt notches **26** as shown in FIG. 2.

(18) To secure the fastening device, the fastening device is first driven through the substrates to be joined. The fastening device may be driven through pre-drilled holes in the substrates. In one embodiment the cap **23** is rigid and pointed so that the fastening device may be pushed or hammered through the substrates. In another embodiment the cap is a drill and the fastening device may be driven through the substrates by rotation of the fastening device. As the fastening device is driven through the substrates, the retaining ring **28** is pressed against the first substrate **40** and is urged out of the retaining ring groove **29** and off of the slide collar **22**. When the retaining ring **28** is removed from the slide collar **22** the stop bars **25** are permitted to rotate in relation to the slide collar **22** and the projections may disengage from the slide bolt notches **26**. With the projections **27** disengaged from the slide bolt notches **26**, the spring **24** biases the slide collar **22** to slide directionally towards the shoulder bolt **10**. As the slide collar **22** slides directionally towards the shoulder bolt **10** the stop bars **25** are biased towards a position perpendicular to the slide bolt **21**. When the slide collar **22** and stop bars **25** emerge on the opposite side of the plurality of substrates, the stop bars **25** are deployed in a position substantially perpendicular to the slide bolt **21**. FIG. 5 shows the fastening device with the stop bars **25** deployed.

(19) Once the stop bars **25** are deployed, the shoulder bolt **10** is rotated causing the slide bolt **21** to move inwardly into the shaft **11** of the shoulder bolt **10**, compressing the substrates between the stop bars **25** and a flange **13** on the shoulder bolt **10**. The shoulder bolt **10** is rotated until a desired torque is reached.

(20) FIG. 1 shows the fastening device fastening a first substrate **40**, a second substrate **41** and third substrate **42**. The retaining ring **28** can be seen between the head **12** and the first substrate **40**. The cap **23**, the slide collar **22** and the stop bars **25** can be seen emerging from the third substrate. The operation of the fastening device is more clearly depicted in FIG. 4. The retaining ring **28** can be seen between the head **12** and the first substrate **40**. As the shoulder bolt **10** is rotated the internally threaded shaft **11** works with the threaded first end of the slide bolt **21** to draw the slide bolt directionally towards the head **12**, compressing the spring **24** between the cap **23** and the slide collar **22** and compressing the substrates between the head **12** and the stop bars **25**.

Additional Embodiments

(21) FIGS. 6-11 illustrate another embodiment of a fastening device or fastener **100**. The fastener **100** can be used to couple with a substrate **150**. The substrate **150** can include one or more substrate layers **151**, **152**. The substrate layers can have widths **W1**, **W2**. An aperture **153** can extend through the substrate **150** from a front side or face **154** to a rear side or face **155**. In some implementations, the aperture **153** can be pre-drilled to allow the fastener **100** to pass through the substrate **150**. The substrate **150** can comprise any material or combination of materials, including building and construction materials such as wooden framing elements, concrete and/or gypsum board or paneling.

(22) The fastener **100** can include a bolt **110**. The bolt **110** can be a conventional bolt such as found in hardware stores and other distributors. The bolt **110** can have a head **112** and a shaft **111**. The shaft **111** can be threaded. The threading and diameter of the shaft **111** can be of any suitable sizing (e.g., conventional sizing such as $\frac{3}{8}$ "-16, #10-24, etc.). The head **112** can be attached to the shaft **111**. The head **112** can have a drive portion shaped to engage with an assembly tool, such as a wrench, socket or screwdriver driver. For example, the head **112** can be externally hexagonal, square or include an internal tool cavity, such as slotted, cruciform, single or multiple square, torx, star or hexagonal. The head **112** can optionally include a flange or be used in combination with a washer, such as a flat washer.

(23) The fastener **100** can further include a biasing element **120** configured to move the fastener **100** relative to the substrate **150**. The biasing element **120** can be or include any suitable

compressible media, such as any type of spring or resilient element. The biasing element **120** can be a permanent or a sacrificial element of the fastener **100**. For example, the biasing element **120** could be or include a polymer element, such as a soft durometer urethane. The biasing element **120** could be configured for removal (e.g., torn away) at any suitable time, such as after the initial installation of the fastener **100**. For example, the biasing element **120** could be removed once the fastener **100** is sufficiently secured that it is unlikely to rotate in response to rotation of the bolt **110**. References to a conical spring **120** herein are for convenience in describing the illustrated embodiment, but can also refer to any other suitable type of biasing element.

(24) In the illustrated arrangement, the biasing element is a cone-shaped compression spring or a conical spring **120**. The conical spring **120** can be mounted on the shaft **111** of the bolt **110**. The conical spring **120** can comprise a narrow end **121** and a wide end **122**. Between the narrow end **121** and the wide end **122**, the overall shape of the conical spring **120** can be generally tapered to form a cone shape or similar shape. The conical spring **120** can be formed of a length of wire **123**. The wire **123** can be coiled to form the tapered shape of the spring **120**. The conical spring **120** can have an uncompressed length and can be compressible to compressed length. The conical spring **120** can exert a force that varies with and/or is proportional to the compression of the spring along its length (e.g., axially between the narrow and tapered ends **121**, **122**). The spring rate of the conical spring **120** can be linear or non-linear.

(25) In some implementations, the conical spring **120** can be compressed substantially or fully into a flattened state. In such a state the coils of the tapered spring of the wire **123** can be nested within each other between the wide end **122** and the narrow end **121** such that the overall spring length is compressed to be substantially the diameter of the wire **123**. In some implementations, the spring **120** can exert its maximum force in the flattened configuration. When assembled over the shaft **111**, the narrow end **121** of the spring **120** can be adjacent to the head **112**. In some implementations, a diameter of the narrow end **121** can be less than the diameter of the head **112** (or a flange thereof). However, in other arrangements, the wide end **122** can be positioned adjacent the head **112**. In such arrangements, the head **112** can have a sufficiently large diameter or cross-sectional dimension to provide a base for the wide end **122** of the conical spring **120** or a washer (e.g., washer **133**) could be employed, among other possible arrangements.

(26) The fastener **100** can further include a slide body **130**. In some implementations, the slide body **130** can be generally cylindrical in shape. The slide body **130** can comprise a first end **131** and a second end **132**. The slide body **130** can be configured to be received within and/or extend through the aperture **153** of the substrate **150**. The slide body **130** can have a diameter. In some implementations, the diameter can be between approximately 0.125" and 1.0". In other implementations, the diameter can be larger or smaller than the range provided here and/or include portions with larger or smaller diameters. A diameter of the aperture **153** can be slightly larger than the diameter of the slide body **130** (e.g., greater than or approximately 0.001 and 0.010 inches larger in some implementations).

(27) The first end **131** of the slide body **130** can include an opening or cavity for receiving the shaft **111** of the bolt **110**. An internal surface of the opening can include threads to threadingly engage with threads of the shaft **111**. A washer **133** can be disposed over the shaft **111** and contact the conical spring **120**. The conical spring **120** can press the washer **133** into contact with the first end **131**. The first end **131** can be generally flat and/or perpendicularly oriented with respect to a longitudinal axis of the slide body **130**.

(28) The second end of **132** of the slide body **130** can comprise or support a toggle **140**. The toggle **140** can comprise a stop link or stop bar. The toggle **140** can be used engage the fastener **100** with the rear face **155** of the substrate **150**. The toggle **140** can include a first end **141** and a second end **142**. The toggle **140** can be coupled with the second end **132** by a pin or pivot **143**. The toggle **140** can be sized and shaped, or otherwise configured, to pass through the aperture **153** in a first configuration and to engage the rear face **155** in a second configuration to thereby prevent passage

through the aperture **153**. The toggle **140** can have a diameter or cross-sectional size that is the same as, or substantially the same as, the slide body **130**.

(29) In the first configuration, the toggle **140** can pass through the aperture **153**. For example, the second end **132** of the slide body **130** can be shaped to be semi-cylindrical in cross-section and the second end **142** of the toggle **140** can be shaped in a complementary manner (e.g., semi-cylindrical in cross-section) to together form a circular cross-section. The first configuration can be formed where the toggle **140** is oriented about the pivot **143** to be axially aligned with slide body **130** to form an outer diameter that fits within the aperture **153**. In other implementations, the toggle **140** can be otherwise compressed in the first configuration (e.g., the toggle **140** can be spring-loaded). In some implementations, the toggle **140** is manually positioned in the first configuration (e.g., by a user). In some implementations, the insertion of the slide body **130** and toggle **140** into the aperture **153** will compress the toggle **140** into the first configuration.

(30) In the second configuration, the toggle **140** can be sized, shaped, oriented or otherwise configured to prevent passage through the aperture **153**. For example, the toggle **140** can have a length TL that is greater than the diameter of the aperture **153**. When inserted through the aperture **153** in the first configuration, the toggle **140** can switch into the second configuration. The length TL defined by a linear surface extending in a lengthwise direction of the toggle **140** in the second configuration (e.g., perpendicular to the aperture **153**) can prevent the toggle **140** and the slide body **130** from being removed from the aperture **153** (i.e., in the direction of the front face **154**).

(31) The toggle **140** can be switched from the first configuration to the second configuration by the force of gravity. For example, the toggle **140** can be eccentrically coupled with the pivot **143** such that the toggle **140** is biased to rotate about the pivot **143** under the force of gravity. As shown in FIG. 8, the first end **141** can have a first length TL1 and the second end **142** can have a second length TL2. The first length TL1 can be greater than the second length TL2 or otherwise configured to be actuated by gravity. For example, the lengths TL1 and TL2 can be configured relative to the overall length TL, shape or center of gravity of the toggle **140** such that the toggle **140** tends to rotate to the second orientation—or out of axial alignment with the slide body—in response to gravity and without any external force sufficient to overcome the gravitational force. In some implementations, the pivot **143** can include a spring (e.g., a torsional spring) to bias the toggle **140** towards or into the second configuration.

(32) In use, the fastener **100** can be assembled with the substrate **150**. This can, for example, couple together the layers **151**, **152**. The spring **120** and the washer **133** can be assembled over the shaft **111** of the bolt **110** and the bolt **110** can be assembled (e.g., threadingly engaged) with the slide body **130**. The toggle **140** can be switched (e.g., manually or by interference with the aperture **153**) into the first configuration and the fastener **100** can be inserted into the substrate **150**. The toggle **140** and/or the second end **132** of the slide body **130** can be inserted into the aperture **153** from the front face **154**, as shown in FIG. 7. The slide body **130** can be advanced within the aperture **153**.

(33) The fastener **100** can be advanced within the aperture **153** until the wide end **122** of the conical spring **120** engages with the front face **151**. The wide end **122** can press against the surface of the front face **152** (e.g., through the optional washer **133**). The narrow end **121** can engage the head **112**. The spring **120** can define a compression length S, the total distance the spring **120** can be compressed. In some implementations, the conical spring **120** can be compressed more than a conventional non-tapered spring having a similar total length because of the conical shape factor of the conical spring **120**.

(34) A lateral force (e.g., from a user) on the head **112** can compress the spring **120** (reduce the compression length S). The compression of the spring (either fully or partially) can advance the toggle **140** out of the aperture **153** on the rear side **155**. In some implementations, the compression length S can be selected such that the pivot **143** can be advanced a clearance length CL out of the aperture **153**. In some implementations, the clearance length CL can be equal to or greater than the second length TL2. In some cases, it may be necessary or desirable for the clearance length CL to

be greater than the second length TL2 such that an entirety of the second end 142 can clear the rear side 155 of the substrate when moving through an arcuate path of motion. Thus, the compression length S of the spring 120 may be substantially equal to or greater than the second length TL2 or the clearance length CL. The compression length S can be selected with consideration of the second length TL2 and shape of the toggle 140 to be sufficient to permit actuation or movement of the toggle 140 from the first configuration to the second configuration. Although the shape of the toggle 140 can influence the exact requirement for the compression length S, it is expected that the compression length S will be approximately equal to or greater than the second length TL2. The toggle 140 can then move towards the second configuration, as shown in-transition in FIG. 8. A release of the force on the head 112 can allow the spring 120 to extend (either partially or fully). The extension of the spring 120 can pull the toggle 140 into engagement with the rear side 155 of the substrate 150, as shown in FIG. 9.

(35) The compression and/or extension of the spring 120 to automatically engage the toggle 140 with the rear side 155 facilitates the installation of the fastener 100 within the substrate 150. The engagement of the toggle 140 with the rear side 155 can enable or facilitate turning of the bolt 110 to tighten the fastener 100 (e.g., by advancing the bolt 110 further within the slide body 130). The engagement of the toggle 140 with the rear side 155 can prevent the slide body 130 from turning with the bolt 110 during tightening of the fastener 100. In some configurations, the force produced by the spring 120 is sufficient to allow the bolt 110 to be tightened without the application of an extraneous force on the slide body 130, which may be difficult to apply. Thus, the spring 120 may hold the toggle 140 against the rear side 155 of the substrate 150 with sufficient force such that friction between the toggle 140 and the rear side 155 (along with any resistance to rotation of the slide body 130 due to engagement with the substrate 155 within the aperture 153) inhibits or prevents rotation of the slide body 130 within the aperture 153. The bolt 110 can thus be tightened as desired by the user to couple the layers 151, 152, as shown in FIG. 10. In some implementations, the bolt 110 is tightened and the conical spring 120 is collapsed. In some implementations, the spring 120 can be fully collapsed to the flattened state (e.g., leaving a thickness only of the wire 123) to enable a firm engagement of the head 112 with the washer 133 and/or the front face 154. The washer 133 can act as a bearing surface for the spring 120 to engage with the front face 154.

(36) The fastener 100 can include a length L. The length L can extend from the first end 131 of the slide body 130 to the pivot 143. The user can select the fastener 100 and/or the configuration of the fastener 100 based on the selection of length L. Thus length L can be based on any of various factors. For example, the length L can be selected by a user based the width of the substrate 150, the compression of the spring 120, the total length of the toggle 140 L and/or the first or second link lengths TL1, TL2. In one implementation, the length L is equal to widths W1 and W2 of the layers 151, 152, plus the clearance length CL (or the second link length TL2), minus a fractional portion (fraction) of one of the widths W1 or W2. In some implementations, the fraction can be between $\frac{1}{8}$ and $\frac{7}{8}$. In one implementation, the fraction is $\frac{1}{4}$. For example, the substrate 150 can include a two-by-four board and a four-by-four (2"×4" and 4"×4" or sanded equivalents). The clearance length CL of the toggle 140 can be 1.0". Following the above formulae using a fraction of $\frac{1}{4}$ of the four-by-four, the length L for the fastener 100 is 6".

(37) The compression length S of the spring 120 preferably is greater than or equal to the clearance length CL. This ensures that the spring 120 can extend far enough at the front side 154 to engage the toggle 140 with the rear side 155 and/or ensures that the spring 120 continues to apply a force to engage the toggle 140 with the rear side 155 (e.g., during tightening) without user intervention between the insertion and tightening steps. The clearance length CL for the purposes of comparison with the compression length S can be adjusted based on the shape and profile of the toggle 140. For example, as shown the clearance length CL can be subtracted by half the diameter of the toggle 140. Other formulae for appropriate lengths of the fastener components also exist and would be within the ordinary skill in view of the description herein.

(38) FIG. 12 illustrates another fastener embodiment, fastener **200**. The fastener **200** can be structured with similar components and used similarly to the fastener **100**, as described above. The fastener **200** can include a bolt **210**, a biasing element (e.g., a conical spring **220**), a slide body **230** and a toggle **240**. The toggle **240** can include first and second stop bars **241**, **242**. The first and second stop bars **241**, **242** can be pivotably engaged with a pivot **243**. The pivot **243** can be engaged with a second end **232** of the slide body **230**. A torsional spring can bias the stop bars **241**, **242**, into a second configuration (e.g., an open position). The second configuration can engage with a back face **255** of a substrate **250**. The stop bars **241**, **242** can be rotated together into a first configuration (e.g., a compressed position). The first configuration can have a diameter that's smaller than an aperture **253** through the substrate **250**.

(39) The fastener **200** can be advanced through the substrate **250** to couple together substrate layers thereof. The toggle **240** can be advanced into the aperture **253** in the first configuration. The spring **220** can engage with a front face **254** of the substrate **250**. Pressure on the bolt **210** can compress the spring **220** and advance the toggle **240** out of the aperture **253**. The toggle **240** can switch into the second configuration (e.g., by expansion of the torsional spring). Thus, a compression length S of the spring **220** preferably is substantially equal to or greater than a length TL of one or both of the stop bars **241**, **242** so that the spring **220** can extend far enough at the front face **254** to engage the toggle **240** with the back face **255**. The compression length S can be selected with consideration of the length TL and shape of the stop bars **241**, **242** to be sufficient to permit actuation or movement of the toggle **240** from the first configuration to the second configuration. Although the shape of the stop bars **241**, **242** can influence the exact requirements, it is expected that the compression length S will be approximately equal to or greater than the length TL .

(40) The first and second stop bars **241**, **242** can be open in the second configuration and prevent removal of the slide body **230** from the aperture **253** by engagement with the back face **255**. The first and second stop bars **241**, **242** can be prevented from expansion away from the slide body **230** by mechanical interference between the stop bars between each other and/or between the second end **232** or the pivot **243** or other means thereby. The spring **220** can extend to automatically engage linear surfaces of respective ones of the stop bars **241**, **242**, that are aligned with one another in a colinear manner, with the back face **255**. The bolt **210** can be tightened by turning with respect to the slide body **230**. The slide body **230** can be held substantially in place by the spring **220** and the toggle **240**. The user can thus tighten the fastener **200**. Tightening of the bolt **210** can include compression of the conical spring **220** to a flattened or substantially flattened state.

(41) The present examples are to be considered illustrative and not restrictive, and the invention is not to be limited to the detail given herein, but may be modified within the scope of the appended claims.

Claims

1. A fastener for attachment to a building construction substrate in which only a front side of the substrate is accessible, comprising: an elongate body comprising a bolt, the elongate body configured to be received within an aperture through the substrate, the elongate body having a first end and a second end, the bolt comprising a head and a threaded shaft; a toggle carried by the elongate body, wherein the toggle is configured to prevent removal of the fastener from the aperture, the toggle comprising a first stop bar having a first linear surface and a second stop bar having a second linear surface, each of the first stop bar and the second stop bar pivotally coupled with the second end of the elongate body, the toggle having a first configuration in which the toggle can advance within the aperture and a second configuration which prevents the fastener from being retracted from the aperture, wherein the first linear surface and the second linear surface are aligned in a colinear manner in the second configuration of the toggle; a spring having a first end and a second end, the spring carried by the bolt; wherein the fastener has an installed position in which

the spring is fully collapsed between the head of the bolt and a front face of the substrate and each of the first linear surface of the first stop bar and the second linear surface of the second stop bar is engaged with a back face of the substrate; wherein the fastener is configured to be converted from an uninstalled position to the installed position by inserting the elongate body into the aperture from the front face of the substrate with the toggle in the first configuration and advancing the elongate body within the aperture until the spring engages with the front face of the substrate; wherein the elongate body is further advanced within the aperture to deform the spring and advance the toggle a clearance length out of the aperture on a back face of the substrate such that the toggle can move to the second configuration; wherein the spring reverts to retract the elongate body within the aperture and wherein a biasing force of the spring is configured to engage each of the first linear surface and the second linear surface of the toggle with the back face of the substrate such that contact between the first linear surface and the second linear surface and the back face inhibits rotation of the toggle relative to the substrate in response to rotation of the bolt and the first stop bar and the second stop bar prevent retraction of the elongate body through the aperture; wherein the bolt is configured to be rotated to collapse the spring and to engage with the front face of the substrate through the spring when the fastener is in the installed position.

2. The fastener of claim 1, wherein the toggle further comprises a torsional spring biases the toggle into the second configuration.

3. The fastener of claim 1, wherein the spring is compressible into a flattened state.

4. The fastener of claim 1, further comprising a washer, the washer disposed on the elongate body between the spring and the front face of the substrate in use, the washer providing a bearing surface for the spring to act against the front face of the substrate.
